1. Finding an asymptotically tight bound for $f(n) = n^2 + 5n - 4$ by using the Θ notation. Justify your answer by demonstrating the constants.

2. Show that $n^2 + 5n - 4 \neq \Theta(n)$ by contradiction.

1. $f(n) = n^2 + 5n - 4 = \Theta(n^2)$

1分 $\Rightarrow c_1 n^2 \le n^2 + 5n - 4 \le c_2 n^2$，其中 $c_1, c_2 > 0$, for all $n \ge n_0$ 1分

同÷n^2 $\Rightarrow c_1 \le 1 + \frac{5}{n} - \frac{4}{n^2} \le c_2$

1分
$c_1 \le 1 + \frac{5}{n} - \frac{4}{n^2}$
$n=1$, $c_1 \le 1+5-4 = 2$
$n=2$, $c_1 \le 1 + \frac{5}{2} - 1 = \frac{5}{2}$
$n=3$, $c_1 \le 1 + \frac{5}{3} - \frac{4}{9} = \frac{20}{9}$
$n=4$, $c_1 \le 1 + \frac{5}{4} - \frac{1}{4} = 2$
$n=5$, $c_1 \le 1 + 1 - \frac{4}{25} = \frac{46}{25}$
$n=6$, $c_1 \le 1 + \frac{5}{6} - \frac{4}{36} = \frac{62}{36}$
⋮
$n=\infty$ $c_1 \le 1 + 0 - 0 = 1$

(一定要告知1(從∞來)

1分 $\Rightarrow c_1$ 取 1

1分
$1 + \frac{5}{n} - \frac{4}{n^2} \le c_2$
$n=1$, $1+5-4 = 2 \le c_2$
$n=2$, $1 + \frac{5}{2} - 1 = \frac{5}{2} \le c_2$
$n=3$, $1 + \frac{5}{3} - \frac{4}{9} = \frac{20}{9} \le c_2$
$n=4$, $1 + \frac{5}{4} - \frac{1}{4} = 2 \le c_2$
$n=5$, $1 + 1 - \frac{4}{25} = \frac{46}{25} \le c_2$
$n=6$, $1 + \frac{5}{6} - \frac{4}{36} = \frac{62}{36} \le c_2$
⋮
$n=\infty$, $1+0-0 = 1 \le c_2$

(至少要寫出c_2取到的n)

1分 $\Rightarrow c_2$ 取 $\frac{5}{2}$

∴取$c_1=1$, $c_2=\frac{5}{2}$, $n_0=1$，則可證 $n^2+5n-4 = \Theta(n^2)$ ※

1分　(c_1、c_2、n_0 只要取的合理皆對)

2.
1分 假設 $n^2+5n-4 = \Theta(n)$ 成立，則c_2、n_0存在

1分
$n^2+5n-4 \le c_2 n$，其中 $c_2 > 0$, for all $n \ge n_0$
同÷n $\Rightarrow n + 5 - \frac{4}{n} \le c_2$
$\Rightarrow n \le c_2 - 5 + \frac{4}{n}$

1分
∵ c_2是常數，n是變數
∴不可能任意n皆成立，跟假設矛盾
得證 $n^2+5n-4 \neq \Theta(n)$ ※

1. Finding an asymptotically tight bound for $f(n) = n^2 + 5n - 4$ by using the Θ notation. Justify your answer by demonstrating the constants.

2. Show that $n^2 + 5n - 4 \neq \Theta(n)$ by contradiction.

1、

let $f(n) = n^2 + 5n - 4$, $g(n) = n^2$

$f(n) = \theta(g(n))$ if there exist positive constants c_1、c_2 and n_0

such that $c_1 g(n) \le f(n) \le c_2 g(n)$ $\forall n \ge n_0$

$\Rightarrow c_1 n^2 \le n^2 + 5n - 4 \le c_2 n^2$

$c_1 n^2 \le n^2 + 5n - 4$

c_1 取 1

$n=1$, $1 \le 2$

$n=2$, $4 \le 10$

$n=3$, $9 \le 20$

$n=4$, $16 \le 32$

⋮

$n^2 + 5n - 4 \le c_2 n^2$

c_2 取 3

$n=1$, $2 \le 3$

$n=2$, $10 \le 12$

$n=3$, $20 \le 27$

$n=4$, $32 \le 48$

⋮

當 $c_1 = 1$, $c_2 = 3$, $n_0 = 1$ 時，存在

$n^2 \le n^2 + 5n - 4 \le 3n^2$

$\forall n \ge 1$

∴ 可證

$n^2 + 5n - 4 = \theta(n^2)$ #

2、

if $n^2 + 5n - 4 = \theta(n)$, then exist positive constants c_2、n_0 s.t.

$n^2 + 5n - 4 \le c_2 n$, $\forall n \ge n_0$

$n^2 + 5n - 4 \le c_2 n \xrightarrow{\text{同除}n} n + 5 - \frac{4}{n} \le c_2$

其中，n 為變數，c_2 為常數。當 $n > c_2$ 時，左式必大於 c_2

這與假設 $\forall n \ge n_0$ 矛盾。

∴ 可證，$n^2 + 5n - 4 \neq \theta(n)$

1. Finding an asymptotically tight bound for $f(n) = n^4 + 3n - 7$ by using the Θ notation. Justify your answer by demonstrating the constants.

藍字為解題的 Tips，可不寫

$f(n) = \Theta(g(n))$ if there exists postitive constants c_1, c_2 and n_0 such that $c_1 \cdot g(n) \le f(n) \le c_2 \cdot g(n)$ for all $n \ge n_0$

$n^4 + 3n - 7 \le c_2 \cdot g(n)$ 取 $g(n) = n^4$

$\le c_2 \cdot n^4$ 取 $c_2 = 2$，通常可取 $f(n)$ 最高項之係數 $+1$ 作為 c_2

$n^4 + 3n - 7 \le 2n^4$

$n^4 - 3n + 7 \ge 0$

$n_0 = n = 1$, $1 - 3 + 7 \ge 0$ (成立)

$\therefore$ 取 $c_2 = 2$, $n_0 = 1$，可證明 $f(n) = n^4 + 3n - 7 = O(n^4)$ — ①　　2分

$n^4 + 3n - 7 \ge c_1 \cdot g(n)$ 取 $g(n) = n^4$

$\ge c_1 \cdot n^4$ 取 $c_1 = 1$，通常可取 $f(n)$ 最高項之係數作為 c_1

$n^4 + 3n - 7 \ge n^4$

$3n - 7 \ge 0$

$n \ge \frac{7}{3}$ (成立)

$\therefore$ 取 $c_1 = 1$, $n_0 = \frac{7}{3}$，可證明 $f(n) = n^4 + 3n - 7 = \Omega(n^4)$ — ②　　2分

根據 ①、②

取 $c_1 = 1$, $c_2 = 2$, $n_0 = \frac{7}{3}$，可證明 $f(n) = n^4 + 3n - 7 = \Theta(n^4)$ #　　1分

2. Show that $3n^3 - 9 \ne O(n^2)$ by contradiction.

可不寫 [若 $3n^3 - 9 = O(n^2)$ 成立，
則代表存在 $3n^3 - 9 \le c \cdot n^2$ for all $n \ge n_0$]

假設 $3n^3 - 9 = O(n^2)$ 成立，

則存在 $3n^3 - 9 \le c \cdot n^2$, $\forall n \ge n_0$

$c \ge 3n - \frac{9}{n^2}$, $\forall n \ge n_0$

c 必須為非常數 (變數)，與假設 c 為常數矛盾

$\therefore 3n^3 - 9 \ne O(n^2)$ #

每一題評分標準 (前 4 點總共最多扣 2 分)：

1. 1分 - c、n_0 為數字，而非範圍
(即將 c、n_0 寫成不等式)
2. 1分 - 過程要寫，列舉如何找出 c、n_0
3. 1分 - 最後答案要明確寫出 c = ？，n_0 = ？
4. 1分 - 取 n = ∞ (n 必須為正整數，$n \ge n_0 \ge 1$)
不要寫在等式中，e.g. $n^2 + 7n + 1 \le 9n^2$
5. 筆墨分數最多 2 分

A 班

1. Finding an asymptotically tight bound for $f(n) = n^4 + 3n - 7$ by using the Θ notation. Justify your answer by demonstrating the constants.

let $f(n) = n^4 + 3n - 7$, $g(n) = n^4$

$f(n) = \theta(g(n))$ if there exist positive constants c_1、c_2 and n_0 such that $c_1 g(n) \le f(n) \le c_2 g(n)$ $\forall n \ge n_0$

$c_1 n^4 \le n^4 + 3n - 7 \le c_2 n^4 \xrightarrow{\text{同除}n^4} c_1 \le 1 + \frac{3}{n^3} - \frac{7}{n^4} \le c_2$

$c_1 \le 1 + \frac{3}{n^3} - \frac{7}{n^4}$ $\qquad$ $1 + \frac{3}{n^3} - \frac{7}{n^4} \le c_2$

$n=1$, $c_1 \le 1 + 3 - 7 = -3$ $\qquad$ $n=1$, $-3 \le c_2$

$n=2$, $c_1 \le 1 + \frac{6}{16} - \frac{7}{16} = \frac{15}{16}$ $\qquad$ $n=2$, $\frac{15}{16} \le c_2$

$n=3$, $c_1 \le 1 + \frac{3}{27} - \frac{7}{81} = \frac{83}{81}$ $\qquad$ $n=3$, $\frac{83}{81} \le c_2$

$n=4$, $c_1 \le 1 + \frac{3}{64} - \frac{7}{256} = \frac{261}{256}$ $\qquad$ $n=4$, $\frac{261}{256} \le c_2$

$\vdots$ $\qquad$ $\vdots$

隨n增長，當$n \to \infty$時，$1 + \frac{3}{n^3} - \frac{7}{n^4}$會趨近於1，s.t. $c_1 \le 1 \le c_2$

$\rightarrow$ c_1取1，n_0取3 $\qquad$ $\rightarrow$ c_2取2

當$c_1 = 1$, $c_2 = 2$, $n_0 = 3$時，存在 $n^4 \le n^4 + 3n - 7 \le 2n^4$, $\forall n \ge 3$

∴可證，$n^4 + 3n - 7 = \theta(n^4)$

2. Show that $3n^3 - 9 \ne O(n^2)$ by contradiction.

if $3n^3 - 9 = O(n^2)$, then exist positive constants c and n_0 such that $3n^3 - 9 \le cn^2$ $\forall n \ge n_0$

$3n^3 - 9 \le cn^2 \xrightarrow{\text{同除}n^2} 3n - \frac{9}{n^2} \le c$

其中，n為變數，c為常數，當$n > c$時，左式必大於c，

這與假設$\forall n \ge n_0$產生矛盾。

∴可證 $3n^3 - 9 \ne O(n^2)$ #

1. Finding an asymptotically tight bound for $f(n) = n^2 + 5n - 7$ by using the O notation. Justify your answer by demonstrating the constants.

藍字為解題的Tips，可不寫

$f(n) = O(g(n))$ if there exists postitive constants c and n_0 such that $f(n) \le c \cdot g(n)$ for all $n \ge n_0$

$n^2 + 5n - 7 \le c \cdot g(n)$ 取 $g(n) = n^2$

$\le c \cdot n^2$ 取 $c = 2$，通常可取 $f(n)$ 最高項之係數 $+1$ 作為 c

$n^2 + 5n - 7 \le 2n^2$

$n^2 - 5n + 7 \ge 0$

$n_0 = n = 1$, $1 - 5 + 7 \ge 0$ (成立)

$\therefore$ 取 $c = 2$, $n_0 = 1$, 可證明 $f(n) = n^2 + 5n - 7 = O(n^2)$ #

2. Show that $3n^3 - 9 \ne \Theta(n^2)$ by contradiction.

可不寫 若 $3n^3 - 9 = \Theta(n^2)$ 成立，則代表存在 $c_1 \cdot n^2 \le 3n^3 - 9 \le c_2 \cdot n^2$ for all $n \ge n_0$

假設 $3n^3 - 9 = O(n^2)$ 成立，

則存在 $3n^3 - 9 \le c_2 \cdot n^2$, $\forall n \ge n_0$

$c_2 \ge 3n - \frac{9}{n^2}$, $\forall n \ge n_0$

c_2 必須為非常數 (變數)，與假設 c_2 為常數矛盾

$\therefore 3n^3 - 9 \underbrace{\ne O(n^2)}_{\text{可不寫}} \ne \Theta(n^2)$ #

B 班

每一題評分標準 (前 4 點總共最多扣 2 分)：

1. 1分 - c、n_0 為數字，而非範圍 (即將 c、n_0 寫成不等式)
2. 1分 - 過程要寫，列舉如何找出 c、n_0
3. 1分 - 最後答案要明確寫出 c = ？，n_0 = ？ 不要寫在等式中，e.g. $n^2 + 7n + 1 \le 9n^2$
4. 1分 - 取 n = ∞ (n 必須為正整數， $n \ge n_0 \ge 1$)
5. 筆墨分數最多 2 分

1. Finding an asymptotically tight bound for $f(n) = n^2 + 5n - 7$ by using the O notation. Justify your answer by demonstrating the constants.

let $f(n) = n^2 + 5n - 7$, $g(n) = n^2$

$f(n) = O(g(n))$ if there exist positive constants c and n_0 such that $f(n) \le c\,g(n)\ \forall n \ge n_0$

$n^2 + 5n - 7 \le cn^2 \xrightarrow{\text{同除}n^2} 1 + \frac{5}{n} - \frac{7}{n^2} \le c$

$n=1$, $1+5-7 \le c$

$n=2$, $\frac{7}{4} \le c$

$n=3$, $\frac{17}{9} \le c$

$n=4$, $\frac{29}{16} \le c$

$n=5$, $\frac{43}{25} \le c$

隨 n 增長，當 $n \to \infty$ 時，$1 + \frac{5}{n} - \frac{7}{n^2}$ 會趨近於 1 s.t. $1 \le c$

$\to$ c 取 2，n_0 取 1

當 $c=2$，$n_0=1$ 時，存在 $n^2+5n-7 \le 2n^2\ \forall n \ge 1$，

$\therefore$ 可證 $n^2+5n-7 = O(n^2)$ #

2. Show that $3n^3 - 9 \neq \Theta(n^2)$ by contradiction.

if $3n^3-9=\Theta(n^2)$, then exist positive constants c_1、c_2 and n_0 such that $c_1 n^2 \le 3n^3-9 \le c_2 n^2 \ \forall n \ge n_0$

$$c_1 n^2 \le 3n^3 - 9 \le c_2 n^2 \xrightarrow{\text{同除} n^2} c_1 \le 3n - \frac{9}{n^2} \le c_2$$

其中，n 為變數，c_2 為常數，當 $n > c_2$ 時，

$3n - \frac{9}{n^2}$ 必大於 c_2，這與假設 $\forall n \ge n_0$ 產生矛盾。

$\therefore$ 可證，$3n^3-9 \neq \Theta(n^2)$

Θ-notation:

if $f(n) = \Theta(g(n))$

there exist positive constants $c_1 \cdot c_2$ and n_0 such that

$c_1 g(n) \le f(n) \le c_2 g(n) \; \forall n \ge n_0$

O-notation:

if $f(n) = O(g(n))$

there exist positive constants c and n_0 such that

$f(n) \le c g(n) \; \forall n \ge n_0$

Ω-notation:

if $f(n) = \Omega(g(n))$

there exist positive constants c and n_0 such that

$c g(n) \le f(n) \; \forall n \ge n_0$